HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Engineered RTL for digital ASIC design flows, focusing on **power efficiency** and **testability** analysis for new semiconductor products.
- Contributed to verification efforts, gaining exposure to industry-standard practices for **Safety** and **Security** within the semiconductor development lifecycle.
- Developed C/C++ code for embedded systems used in low-power edge sensing applications, supporting product innovation in consumer wearables.
- Participated in global project teams, applying state-of-the-art methodologies to assignments within the **Analog Design** and **Digital Design** domains.
- Wrote Python scripts to automate data extraction and analysis for design verification reports, improving data processing efficiency by **15%**.
- Collaborated with senior engineers on **Verilog** code reviews and debugging of digital design modules, ensuring adherence to design specifications.
- Containerized development environments using **Docker** and managed deployments on **AWS EC2** instances for team-wide access to design tools.

SKILLS

Hardware Design – Verilog, SystemVerilog, RTL Development, Synthesis, Analysis

Programming Languages – C/C++, Python, PERL

Semiconductor Domains – Analog IC Design, Digital ASIC Design, Verification, Safety, Security, Mixed Signal Verification

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Software & Scripting – Python, Shell Scripting

PROJECTS

Digital ASIC Design Flow Automation

- Developed Python scripts to automate the setup and execution of RTL synthesis and static timing analysis for digital ASIC designs, reducing manual setup time by **4 hours per design cycle**.
- Integrated scripting with existing **Verilog** codebases, enabling automated generation of design reports and compliance checks against power efficiency targets.
- Managed code versions using **Git** and facilitated collaboration within a small team on the automation scripts and design verification components.

Analog Circuit Simulation and Analysis Tool

- Built a Python application to parse simulation results from analog circuit designs, generating summary reports and visualizations of key performance indicators.
- Applied principles of **Analog IC Knowledge** to interpret simulation outputs and identify potential areas for design improvement, contributing to a **10% reduction** in simulation debugging time.
- Developed data processing logic using **NumPy** and visualization components with **Matplotlib** to present complex analog behavior in an understandable format.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)